University of North Georgia

Computer Science & Cyber Security

CSCI 3250 Computer Security

General policies: Work in team of two students. Working as a solo cuts 10 points. Each student submits the same file.

Presenting Research on a Computer Security Topic.

Description: In this project, students will conduct research on a selected topic in computer security and present their findings in a formal presentation. This project aims to deepen students' understanding of complex security issues, improve their ability to analyze and communicate technical information, and build skills in creating organized, professional presentations. Students will learn to convey cybersecurity concepts effectively to both technical and non-technical audiences.

Specific Learning Outcomes:

1. Developing Research Skills in Computer Security:
    - Students will explore a specific topic of their interest within computer security (please check list below), performing research using credible academic and industry sources to gather current information and findings.
2. Analyzing and Synthesizing Information:
    - Students will critically analyze research materials, summarize key points, and synthesize the information to present a coherent and comprehensive understanding of their topic. They will learn to differentiate between credible sources and understand recent developments and challenges within their chosen area.
3. Creating a Clear and Professional Presentation:
    - Students will develop skills in creating a professional presentation, organizing content logically, and using visual aids to enhance understanding. They will learn to structure a presentation effectively, balancing technical detail with clear explanations for accessibility.

Overall Project Details

You will create a team of two/three members and select a relevant Computer Security topic/problem of your interest – see list at the end- and then present your findings to the class. The goal of the presentation is to identify why your topic is relevant to the Computer Security industry today. The presentation must be interesting and professional. You and your team member will work on and turn in the same research together, but each person must submit the same copy of the research presentation into D2L to receive credit.

- Use the presentation plan below as your framework to guide you through each section of your research presentation. The plan listed below uses the example of Quantum Computing and Cryptography as a topic. Replace Quantum Computing with your topic to properly answer each item.
- This framework will serve as your blueprint to complete the research and presentation with a consistent flow of information about your topic. To receive full credit, you must include each of the items listed in the presentation plan. You are welcome to go beyond this list, but you must satisfy each of the items below at a minimum.
- Provide the presentation plan as seen below with at least 3 sentences for each item. The 3 sentences should provide precise details to address each item. Do not submit generic, vague, or unclear sentences. This part of the project is where you begin to finalize details about your overall project. I want you to provide those specific details for each of these items such that it is very clear what are going to accomplish with your research topic.
- Presentation Plan:
    - Team Introduction
    - What is Quantum Computing?
    - How does it relate to Computer Security?
    - Why should you be aware of Quantum Computing as it relates to computer security?
    - Has any security protocol or algorithm been broken by Quantum Computing?
    - Quantum Computing demonstration.
    - Ways to protect data in the Quantum Computing era.

    - Books, videos, and tutorials to help you understand Quantum Computing and its relation to computer security.
    - Conclusion
- Must provide at least 4 separate supporting references.
    - Books, journals, conference presentation, research papers, etc.

List of topics to choose from:

AI & Machine Learning Security

1. Using AI for intrusion detection systems
2. Adversarial attacks against machine learning models
3. AI-generated phishing detection techniques
4. Malware classification using machine learning

Cryptography & Emerging Threats

5. Post-quantum cryptography and its impact on current systems
6. Security analysis of hash functions in authentication
7. Blockchain-based identity management
8. Lightweight cryptography for IoT devices

Network & Systems Security

9. Zero-trust architecture in enterprise networks
10. Detection of distributed denial-of-service (DDoS) attacks
11. Network anomaly detection using AI
12. Security challenges in 5G networks

Web & Application Security

13. Cross-Site Scripting (XSS) prevention techniques
14. Secure software development lifecycle practices
15. API security vulnerabilities in modern web apps
16. Software supply-chain attacks and mitigation

IoT, Cloud, and Emerging Technologies

17. IoT device security vulnerabilities and countermeasures
18. Cloud security misconfiguration analysis
19. Edge computing security challenges

20. Security risks in smart city infrastructure

Grading Criteria:

1. Submitting the research presentation as a power point/pdf file is 10 points.
2. Presenting to classmates. 20 points.
3. At least 4 references as requested. 10 points.
4. Presentation includes a demo – YouTube video. 10 points.
5. Each student must submit the same pdf files in D2L. If a team member does not submit the pdf file, 30 points will be cut from the whole project for that student.
6. Different grading criteria applies when responsibilities are not completely fulfilled. A student that didn't complete his/her part will receive zero.
7. 10 points will be cut if there is no evaluation of team members as requested.
8. The project will be graded for completeness. An incomplete project will receive credit according with what was accomplish.
9. If a student doesn't attend the team's presentation 30 points will be cut.

<u>Participation</u>

- Students are to remain professional and give full attention to presentations. Student's roster will be called.